

Ex.no:4	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

Algorithm:

1. Import the required Pandas, NumPy and Matplotlib libraries.
2. Read the stud_data.csv file using Pandas.
3. Display the dataset and inspect the first few records.
4. Find the number of rows and columns and display the data types.
5. Generate the statistical summary of the dataset.
6. Filter the students based on marks and attendance conditions.
7. Sort the student records based on marks and attendance.
8. Group the data based on department and calculate average marks and attendance.
9. Calculate mean, median, minimum, maximum and standard deviation of marks.
10. Calculate the correlation between marks and attendance.
11. Create bar chart, line chart, scatter plot, histogram and box plot using Matplotlib.
12. Display the results and interpret the obtained data.

Program:

```
# Import libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Read dataset
```

```
df = pd.read_csv("stud_data.csv")
```

```
# Display dataset
```

```
print(df)
```

```
# Display first five records
```

```
print(df.head())
```

```
# Display shape
```

```
print(df.shape)
```

```
# Display data types
```

```
print(df.dtypes)
```

```
# Display statistical summary
```

```
print(df.describe())
```

```
# Filter students with marks greater than 80
```

```
print(df[df["Marks"] > 80])
```

```
# Filter students with attendance greater than 90
```

```
print(df[df["Attendance"] > 90])
```

```
# Sort students by marks
```

```
print(df.sort_values("Marks", ascending=False))
```

```
# Sort students by attendance
```

```
print(df.sort_values("Attendance", ascending=False))
```

```
# Department-wise average marks
```

```
print(df.groupby("Department")["Marks"].mean())
```

```
# Department-wise average attendance
```

```
print(df.groupby("Department")["Attendance"].mean())
```

```
# Calculate mean
```

```
print("Mean Marks:", df["Marks"].mean())
```

```
# Calculate median
```

```
print("Median Marks:", df["Marks"].median())
```

```
# Calculate minimum
```

```
print("Minimum Marks:", df["Marks"].min())
```

```
# Calculate maximum

print("Maximum Marks:", df["Marks"].max())


# Calculate standard deviation

print("Standard Deviation:", df["Marks"].std())


# Display correlation matrix

print(df[["Marks", "Attendance"]].corr())


# Display correlation between marks and attendance

print("Correlation:", df["Marks"].corr(df["Attendance"]))


# Bar chart

plt.bar(df["Name"], df["Marks"])

plt.title("Student Marks")

plt.xlabel("Name")

plt.ylabel("Marks")

plt.show()


# Line chart

plt.plot(df["Name"], df["Attendance"], marker="o")

plt.title("Student Attendance")

plt.xlabel("Name")

plt.ylabel("Attendance")
```

```
plt.show()
```

```
# Scatter plot
```

```
plt.scatter(df["Attendance"], df["Marks"])
```

```
plt.title("Marks vs Attendance")
```

```
plt.xlabel("Attendance")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

```
# Histogram
```

```
plt.hist(df["Marks"], bins=5)
```

```
plt.title("Marks Distribution")
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Number of Students")
```

```
plt.show()
```

```
# Box plot
```

```
plt.boxplot(df["Marks"])
```

```
plt.title("Marks Box Plot")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

Output:

	Roll No	Name	Department	Marks	Attendance
0	101	Arun	CSE	85	92
1	102	Bala	IT	78	88
2	103	Divya	CSE	95	96
3	104	Priya	AI&DS	90	94
4	105	Ravi	IT	72	85
5	106	Karthik	AI&DS	88	91

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(6, 5)

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

	Roll No	Marks	Attendance
count	6.000000	6.000000	6.00
mean	103.500000	84.666667	91.00
std	1.870829	8.382521	4.00
min	101.000000	72.000000	85.00
25%	102.250000	79.750000	88.75
50%	103.500000	86.500000	91.50
75%	104.750000	89.500000	93.50
max	106.000000	95.000000	96.00

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Department

AI&DS 89.0

CSE 90.0

IT 75.0

Name: Marks, dtype: float64

Department

AI&DS 92.5

CSE 94.0

IT 86.5

Name: Attendance, dtype: float64

Mean Marks: 84.66666666666667

Median Marks: 86.5

Minimum Marks: 72

Maximum Marks: 95

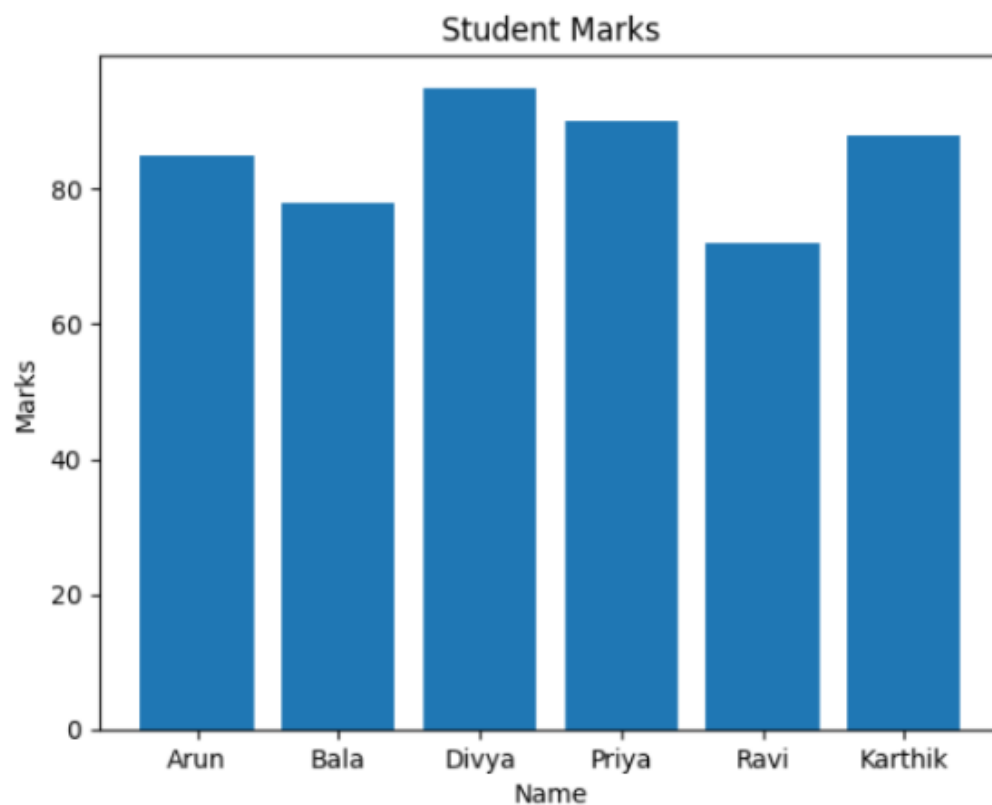
Standard Deviation: 8.382521498133285

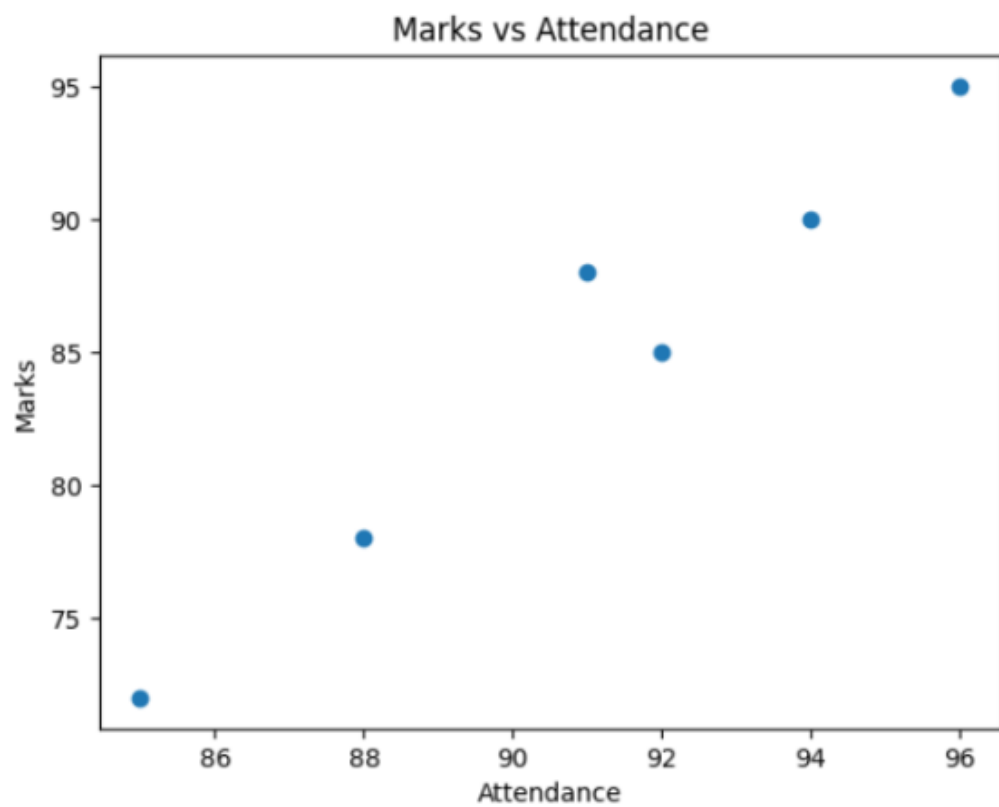
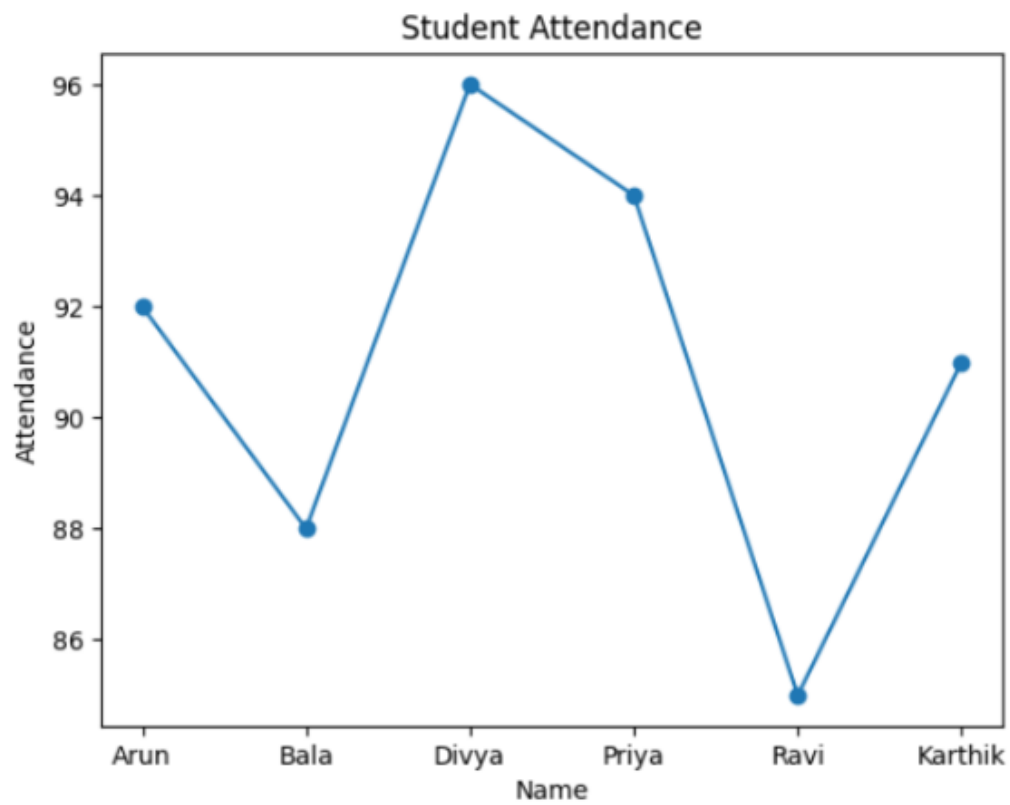
	Marks	Attendance
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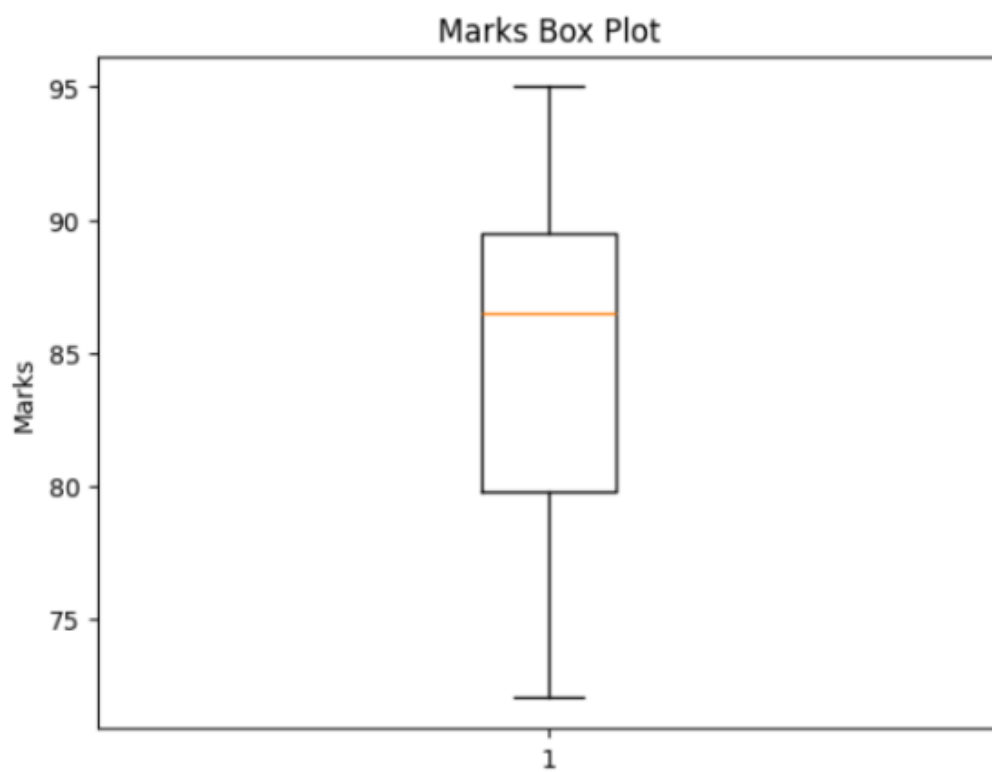
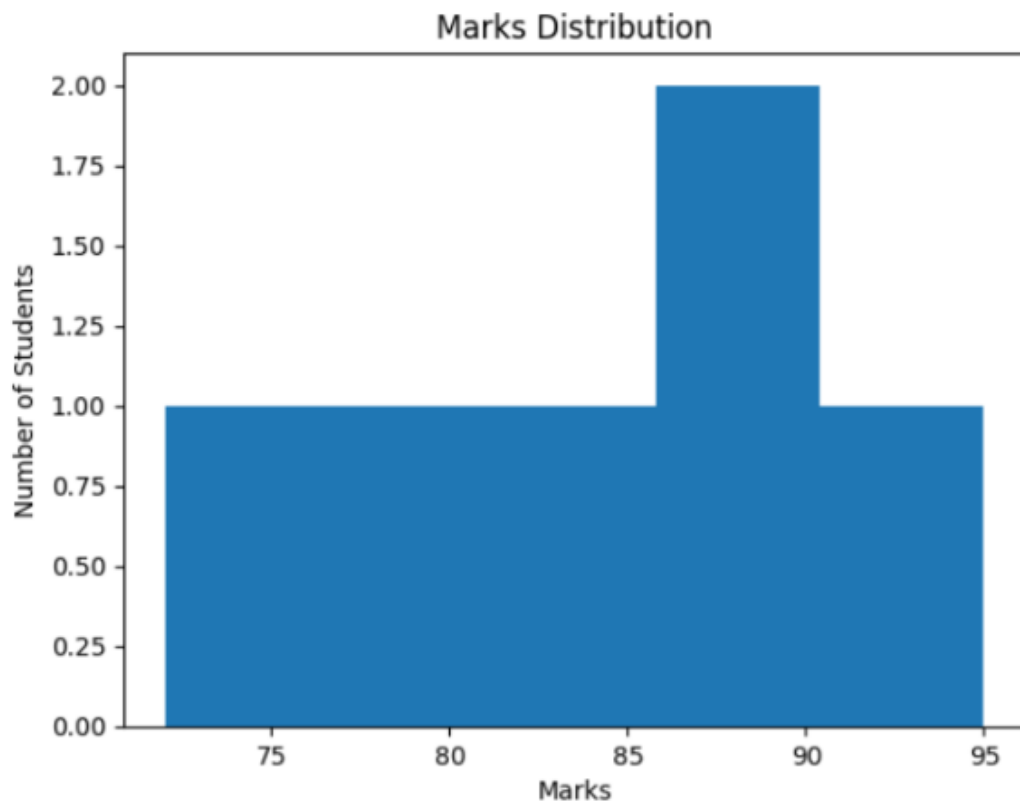
Marks	1.000000	0.978226
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Attendance	0.978226	1.000000
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Correlation: 0.9782259433305447







Result: